Money, Government Deficits, and Inflation

Money Demand:

$$\frac{M_t^d}{P_t} = \Phi(\overset{-}{i}_t, \overset{+}{Y}_t)$$

where

$$1 + i_t = (1 + r_t)\frac{\mathbb{E}_t P_{t+1}}{P_t}$$

For simplicity, assume:

$$\begin{aligned} Y_t &= Y \quad \forall t \\ r_t &= r \quad \forall t \end{aligned}$$

Then:

$$\frac{M_t^d}{P_t} = \Psi\left(\frac{\mathbb{E}_t \overset{-}{P}_{t+1}}{P_t}\right)$$

where $\Psi\left(\frac{\mathbb{E}_t P_{t+1}}{P_t}\right) \equiv \Phi\left((1+r)\frac{\mathbb{E}_t P_{t+1}}{P_t} - 1, Y\right)$.

For simplicity, we assume that the function Ψ is linear:

$$\frac{M_t^d}{P_t} = c - b\frac{\mathbb{E}_t P_{t+1}}{P_t}$$

with $c > 0$ and $b > 0$.

Equilibrium requires:

$$M_t^d = M_t$$

where M_t is the money supply (in nominal terms).

Then:

$$\frac{M_t}{P_t} = c - b\frac{\mathbb{E}_t P_{t+1}}{P_t}$$

From the previous expression we get:

$$P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$$

where $\alpha \equiv \frac{b}{c} > 0$.

The expression above is a first-order stochastic linear difference equation in the price level.

Assume:

$$\alpha < 1$$

Then, we can solve $P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$ forward:

$$\begin{aligned}
P_t &= \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1} \\
&= \frac{M_t}{c} + \alpha \mathbb{E}_t \{ \frac{M_{t+1}}{c} + \alpha \mathbb{E}_{t+1} P_{t+2} \} \\
&= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \mathbb{E}_{t+1} P_{t+2} \\
&= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t P_{t+2} \quad \text{by LIE} \\
&= \frac{M_t}{c} + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t \{ \frac{M_{t+2}}{c} + \alpha \mathbb{E}_{t+2} P_{t+3} \} \\
&= \frac{1}{c} M_t + \frac{\alpha}{c} \mathbb{E}_t M_{t+1} + \frac{\alpha^2}{c} \mathbb{E}_t M_{t+2} + \alpha^3 \mathbb{E}_t \mathbb{E}_{t+2} P_{t+3} \\
&= \frac{1}{c} (M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2}) + \alpha^3 \mathbb{E}_t P_{t+3}
\end{aligned}$$

If we keep iterating forward we get:

$P_t = \frac{1}{c}(M_t + \alpha \mathbb{E}_t M_{t+1} + \alpha^2 \mathbb{E}_t M_{t+2} + ... + \alpha^n \mathbb{E}_t M_{t+n}) + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$

Then:

$$P_t = \frac{1}{c} \sum_{i=0}^{n} \alpha^i \mathbb{E}_t M_{t+i} + \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Letting $n \to \infty$:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} + \lim_{n \to \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

assuming that $\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \equiv \lim_{n \to \infty} \sum_{i=0}^{n} \alpha^i \mathbb{E}_t M_{t+i}$ is finite.

Fundamental Solution

Notation: we use a $*$ to denote this particular solution.

Suppose the price level satisfies:

$$\lim_{n\to\infty} \alpha^n \mathbb{E}_t P_{t+n} = 0$$

Then, the previous expression for P_t becomes:

$$P_t^* = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \tag{1}$$

The expression above shows that the price level at t depends on the money supply expected to prevail from t into the indefinite future.

According to previous equation, if government deficits are to influence the price level path, it can only be through their effect on the expected path of base money. However, the government deficit and path of base money are not rigidly linked in any immutable way. The reason is that the government can, at least to a point, borrow by issuing interest-bearing government debt, and so need not necessarily issue base money to cover its deficit. To see this more precisely, we need to write down the budget constraint of the consolidated government:

$$G_t + r_{t-1} D_{t-1} = T_t + (D_t - D_{t-1}) + \frac{M_t - M_{t-1}}{P_t} \quad (2)$$

Under the system formed by equations (1) and (2), the inflationary consequences of government deficits depend sensitively on the government's strategy for servicing the debt that it issues.

We consider first a regime in which government deficits have no effects on the rate of inflation. In this regime, the government always finances its entire deficit or surplus by issuing or retiring interest-bearing government debt. Then:

$$M_t - M_{t-1} = 0 \quad \forall t$$

and

$$G_t + r_{t-1} D_{t-1} = T_t + (D_t - D_{t-1}) \quad \forall t$$

From $M_t - M_{t-1} = 0 \ \forall t$ we get:

$$M_t = M \quad \forall t$$

Then, (1) becomes:

$$\begin{aligned} P_t^* &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \\ &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M \\ &= \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i M \\ &= \frac{M}{c} \sum_{i=0}^{\infty} \alpha^i \\ &= \frac{M}{c} [1 + \alpha + \alpha^2 + ...] \\ &= \frac{M}{c} \frac{1}{1-\alpha} \quad \text{since } \alpha \in (0,1) \end{aligned}$$

That is:

$$P_t^* = \frac{1}{(1-\alpha)c} M$$

Hence, P_t^* is constant over time and proportional to M.

Also, from $G_t + r_{t-1}D_{t-1} = T_t + (D_t - D_{t-1})$ we get:

$$D_{t-1} = \frac{T_t - G_t}{1 + r_{t-1}} + \frac{D_t}{1 + r_{t-1}}$$

Shifting the expression above one period ahead:

$$D_t = \frac{T_{t+1} - G_{t+1}}{1 + r_t} + \frac{D_{t+1}}{1 + r_t}$$

Since $\mathbb{E}_t D_t = D_t$, the expression above gives:

$$D_t = \mathbb{E}_t \left\{ \frac{T_{t+1} - G_{t+1}}{1 + r_t} + \frac{D_{t+1}}{1 + r_t} \right\}$$

Iterating forward on the previous expression we get:

$$\begin{aligned} D_t &= \mathbb{E}_t \left\{ \tfrac{T_{t+1} - G_{t+1}}{1+r_t} + \tfrac{1}{1+r_t} \left(\tfrac{T_{t+2} - G_{t+2}}{1+r_{t+1}} + \tfrac{D_{t+2}}{1+r_{t+1}} \right) \right\} \\ &= \mathbb{E}_t \left\{ \tfrac{T_{t+1} - G_{t+1}}{1+r_t} + \tfrac{T_{t+2} - G_{t+2}}{(1+r_t)(1+r_{t+1})} + \tfrac{D_{t+2}}{(1+r_t)(1+r_{t+1})} \right\} \end{aligned}$$

Then:

$$D_t = \mathbb{E}_t \left\{ \begin{array}{l} \frac{T_{t+1} - G_{t+1}}{1+r_t} + \frac{T_{t+2} - G_{t+2}}{(1+r_t)(1+r_{t+1})} \\ + \frac{1}{(1+r_t)(1+r_{t+1})} \left(\frac{T_{t+3} - G_{t+3}}{1+r_{t+2}} + \frac{D_{t+3}}{1+r_{t+2}} \right) \end{array} \right\}$$

$$= \mathbb{E}_t \left\{ \begin{array}{l} \frac{T_{t+1} - G_{t+1}}{1+r_t} + \frac{T_{t+2} - G_{t+2}}{(1+r_t)(1+r_{t+1})} + \frac{T_{t+3} - G_{t+3}}{(1+r_t)(1+r_{t+1})(1+r_{t+2})} \\ + \frac{D_{t+3}}{(1+r_t)(1+r_{t+1})(1+r_{t+2})} \end{array} \right\}$$

$$= \mathbb{E}_t \left\{ \begin{array}{l} \frac{T_{t+1} - G_{t+1}}{1+r_t} + \frac{T_{t+2} - G_{t+2}}{(1+r_t)(1+r_{t+1})} + ... + \frac{T_{t+n+1} - G_{t+n+1}}{(1+r_t)(1+r_{t+1})...(1+r_{t+n})} \\ + \frac{D_{t+n+1}}{(1+r_t)(1+r_{t+1})...(1+r_{t+n})} \end{array} \right\}$$

Then:

$$D_t = \mathbb{E}_t \sum_{j=0}^{n} \frac{1}{R_{tj}} (T_{t+j+1} - G_{t+j+1}) + \mathbb{E}_t \left\{ \frac{1}{R_{tn}} D_{t+n+1} \right\}$$

where $R_{tj} \equiv (1 + r_t)(1 + r_{t+1})...(1 + r_{t+j})$.

Remark: with r_t constant we get $R_{tj} = (1 + r)^{j+1}$.

Now let $n \longrightarrow \infty$ and set $\lim_{n\to\infty} E_t\left\{\frac{1}{R_{tn}} D_{t+n+1}\right\} = 0$. Then:

$$D_t = \mathbb{E}_t \sum_{j=0}^{\infty} \frac{T_{t+j+1} - G_{t+j+1}}{R_{tj}}$$

In this regime, a positive value of interest-bearing government debt signals a stream of future government budgets that is in surplus in the present value sense indicated by the previous equation.

Substituting the expression above into $D_{t-1} = \frac{T_t - G_t}{1+r_{t-1}} + \frac{D_t}{1+r_{t-1}}$ we get:

$$(1 + r_{t-1}) D_{t-1} = (T_t - G_t) + \mathbb{E}_t \sum_{j=0}^{\infty} \frac{T_{t+j+1} - G_{t+j+1}}{R_{tj}}$$

In this regime, government deficits have no effects on the price path because they are permitted to have no effects on the path of base money. For the path of base money to be unaffected by government deficits, it is necessary that government (primary) deficits be temporary and be accompanied by exactly offsetting future government surpluses.

There are alternative debt-servicing strategies under which government deficits *are* inflationary. To take an example at the opposite pole from the previous regime, consider the following rule:

$$D_t = 0 \quad \forall t$$

Then, equation (2) becomes

$$G_t - T_t = \frac{M_t - M_{t-1}}{P_t}$$

According to this rule, deficits are always to be financed entirely by issuing additional base money, with interest-bearing government debt never being issued. In this regime, the time path of government deficits affects the time path of the price level in a rigid and immediate way that is described by the previous budget constraint and the equation for P_t^*. Under this regime it is possible for the government budget to be persistently in deficit, within limits imposed by the budget constraint and the money-demand equation. Deficits need not be temporary.

As a simple example, suppose that the primary deficit is constant:

$$G_t - T_t = \Delta \quad \forall t$$

This requires a constant seigniorage:

$$\frac{M_t - M_{t-1}}{P_t} = \Delta \quad \forall t$$

We guess that, in this case, there is an equilibrium with a constant growth of the money supply:

$$M_{t+1} = (1+\mu)M_t \quad \forall t$$

Then:

$$M_{t+i} = (1+\mu)^i M_t$$

Substituting into the expression for P_t^* :

$$\begin{aligned} P_t^* &= \frac{1}{c}\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} \\ &= \frac{1}{c}\sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t \left\{(1+\mu)^i M_t\right\} \\ &= \frac{M_t}{c}\sum_{i=0}^{\infty} [\alpha(1+\mu)]^i \end{aligned}$$

Then, assuming $\alpha(1+\mu) < 1$:

$$P_t^* = \frac{M_t}{c[1-\alpha(1+\mu)]}$$

Since P_t^* is proportional to M_t, we get:

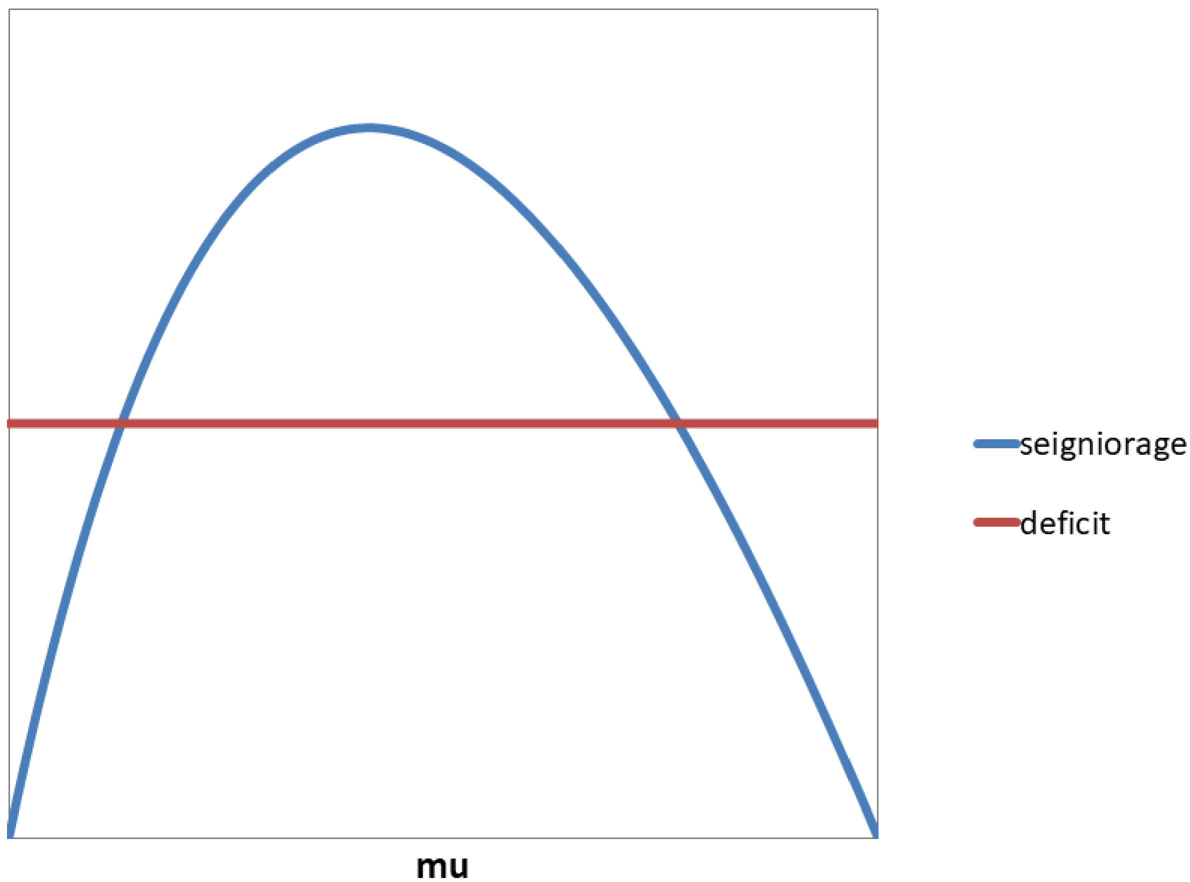
$$\pi_t = \mu \quad \forall t$$

where $\pi_t \equiv \frac{P_t - P_{t-1}}{P_{t-1}}$ is the inflation rate at t.

To determine the equilibrium value of μ we need to go back to the budget constraint:

$$\begin{aligned}
\Delta &= \frac{M_t - M_{t-1}}{P_t} \\
&= \frac{M_t - M_{t-1}}{M_{t-1}} \frac{M_{t-1}}{P_t} \\
&= \mu \frac{M_{t-1}}{M_t} \frac{M_t}{P_t} \\
&= \frac{\mu}{1+\mu} c[1-\alpha(1+\mu)]
\end{aligned}$$

From the expression above we can solve for μ. In this particular case we get a quadratic equation than can be solved in closed form, and we obtain multiple equilibria (assuming the equilibria exist). See the figure below.


seigniorage
deficit
mu

Appendix I

Budget Constraint of the Consolidated Government

Treasury (T)

Budget constraint:

$$G_t^{\mathtt{T}} + r_{t-1} D_{t-1}^{\mathtt{T}} = T_t + (D_t^{\mathtt{T}} - D_{t-1}^{\mathtt{T}}) + V_t$$

Central Bank (CB)

Balance Sheet:

$D_t^{\mathtt{T,CB}}$	M_t/P_t
	$D_t^{\mathtt{CB}}$
	NW_t

Budget constraint:

$$G_t^{\mathtt{CB}} + r_{t-1} D_{t-1}^{\mathtt{CB}} + V_t + (D_t^{\mathtt{T,CB}} - D_{t-1}^{\mathtt{T,CB}}) = r_{t-1} D_{t-1}^{\mathtt{T,CB}} + (D_t^{\mathtt{CB}} - D_{t-1}^{\mathtt{CB}}) + \frac{M_t - M_{t-1}}{P_t}$$

Consolidated Government

Combining the budget constraints of T and CB:

$$\begin{aligned} & G_t^{\mathtt{T}} + r_{t-1} D_{t-1}^{\mathtt{T}} + G_t^{\mathtt{CB}} + r_{t-1} D_{t-1}^{\mathtt{CB}} + V_t + (D_t^{\mathtt{T,CB}} - D_{t-1}^{\mathtt{T,CB}}) \\ = \; & T_t + (D_t^{\mathtt{T}} - D_{t-1}^{\mathtt{T}}) + V_t + r_{t-1} D_{t-1}^{\mathtt{T,CB}} + (D_t^{\mathtt{CB}} - D_{t-1}^{\mathtt{CB}}) + \frac{M_t - M_{t-1}}{P_t} \end{aligned}$$

Then:

$$\begin{aligned} & (G_t^{\mathtt{T}} + G_t^{\mathtt{CB}}) + r_{t-1}(D_{t-1}^{\mathtt{T}} - D_{t-1}^{\mathtt{T,CB}} + D_{t-1}^{\mathtt{CB}}) \\ = \; & T_t + (D_t^{\mathtt{T}} - D_t^{\mathtt{T,CB}} + D_t^{\mathtt{CB}}) - (D_{t-1}^{\mathtt{T}} - D_{t-1}^{\mathtt{T,CB}} + D_{t-1}^{\mathtt{CB}}) + \frac{M_t - M_{t-1}}{P_t} \end{aligned}$$

Then:

$$G_t + r_{t-1} D_{t-1} = T_t + D_t - D_{t-1} + \frac{M_t - M_{t-1}}{P_t}$$

where

$$\begin{aligned} G_t & \equiv G_t^{\mathtt{T}} + G_t^{\mathtt{CB}} \\ D_t & \equiv D_t^{\mathtt{T}} - D_t^{\mathtt{T,CB}} + D_t^{\mathtt{CB}} \end{aligned}$$

Appendix II:

Solution with Bubbles

The equation we want to solve is

$$P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$$

with $\alpha \in (0, 1)$.

We have already shown that:

$$P_t = \frac{1}{c} \sum_{i=0}^{\infty} \alpha^i \mathbb{E}_t M_{t+i} + \lim_{n \to \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1}$$

Now we want a solution with

$$\lim_{n \to \infty} \alpha^{n+1} \mathbb{E}_t P_{t+n+1} \neq 0$$

We propose the following candidate:

$$P_t = P_t^* + B_t$$

where P_t^* is the fundamental solution and B_t is an unspecified random variable (a bubble).

Now we check under what conditions (if any) the expression above is a solution.

For $P_t = P_t^* + B_t$ to be a solution it must satisfy our original equation, $P_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}$. Hence, we need:

$$P_t^* + B_t = \frac{M_t}{c} + \alpha \mathbb{E}_t \{P_{t+1}^* + B_{t+1}\}$$

Then:

$$P_t^* + B_t = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}^* + \alpha \mathbb{E}_t B_{t+1}$$

But we know the fundamental solution satisfies the original equation:

$$P_t^* = \frac{M_t}{c} + \alpha \mathbb{E}_t P_{t+1}^*$$

Combining the previous two expressions we get:

$$B_t = \alpha \mathbb{E}_t B_{t+1}$$

Hence, $P_t = P_t^* + B_t$ will be a solution as long as the random variable B_t (the bubble) satisfies the condition above.

From the previous expression we get:

$$\mathbb{E}_t B_{t+1} = \frac{1}{\alpha} B_t$$

Notice that $\frac{1}{\alpha} > 1$, since $\alpha \in (0, 1)$.

We have:

$$\mathbb{E}_{t+1} B_{t+2} = \frac{1}{\alpha} B_{t+1}$$

Then:

$$
\begin{aligned}
\mathbb{E}_t \mathbb{E}_{t+1} B_{t+2} &= \mathbb{E}_t \{ \frac{1}{\alpha} B_{t+1} \} \\
\mathbb{E}_t B_{t+2} &= \frac{1}{\alpha} \mathbb{E}_t B_{t+1} \quad \text{by LIE} \\
&= \frac{1}{\alpha} \frac{1}{\alpha} B_t \\
&= \frac{1}{\alpha^2} B_t
\end{aligned}
$$

And:

$$
\begin{aligned}
\mathbb{E}_{t+2} B_{t+3} &= \frac{1}{\alpha} B_{t+2} \\
\mathbb{E}_t \mathbb{E}_{t+2} B_{t+3} &= \frac{1}{\alpha} \mathbb{E}_t B_{t+2} \\
\mathbb{E}_t B_{t+3} &= \frac{1}{\alpha} \mathbb{E}_t B_{t+2} \quad \text{by LIE} \\
\mathbb{E}_t B_{t+3} &= \frac{1}{\alpha} \frac{1}{\alpha^2} B_t \\
\mathbb{E}_t B_{t+3} &= \frac{1}{\alpha^3} B_t
\end{aligned}
$$

Keep going to get:

$$\mathbb{E}_t B_{t+n} = \frac{1}{\alpha^n} B_t$$

Then:

$$\begin{aligned}
\lim_{n\to\infty} \mathbb{E}_t B_{t+n} &= \lim_{n\to\infty} \frac{1}{\alpha^n} B_t \\
&= B_t \lim_{n\to\infty} \frac{1}{\alpha^n} \\
&= \begin{cases} +\infty & \text{if} \quad B_t > 0 \\ -\infty & \text{if} \quad B_t < 0 \end{cases}
\end{aligned}$$

Hence, the bubble is explosive.

Just to be sure, we check that $\lim_{n\to\infty} \alpha^{n+1}\mathbb{E}_t P_{t+n+1} \neq 0$ whenever there is a bubble ($B_t \neq 0$).

Using $P_t = P_t^* + B_t$ we get:

$$
\begin{aligned}
\lim_{n\to\infty} \alpha^{n+1}\mathbb{E}_t P_{t+n+1} &= \lim_{n\to\infty} \alpha^{n+1}\mathbb{E}_t\{P_{t+n+1}^* + B_{t+n+1}\} \\
&= \underbrace{\lim_{n\to\infty} \alpha^{n+1}\mathbb{E}_t P_{t+n+1}^*}_{=0} \\
&\quad + \lim_{n\to\infty} \alpha^{n+1}\mathbb{E}_t B_{t+n+1} \\
&= \lim_{n\to\infty} \alpha^{n+1}\mathbb{E}_t B_{t+n+1} \\
&= \lim_{n\to\infty} \alpha^{n+1}\frac{1}{\alpha^{n+1}} B_t \\
&= \lim_{n\to\infty} B_t \\
&= B_t \\
&\neq 0 \quad \text{whenever } B_t \neq 0
\end{aligned}
$$

Example

Suppose:

$$M_t = M$$

and $\{B_t\}$ satisfies

$$B_{t+1} = \frac{1}{\alpha} B_t + \eta_{t+1}$$

where $\{\eta_{t+1}\}$ is a zero-mean iid sequence of shocks.

Notice that

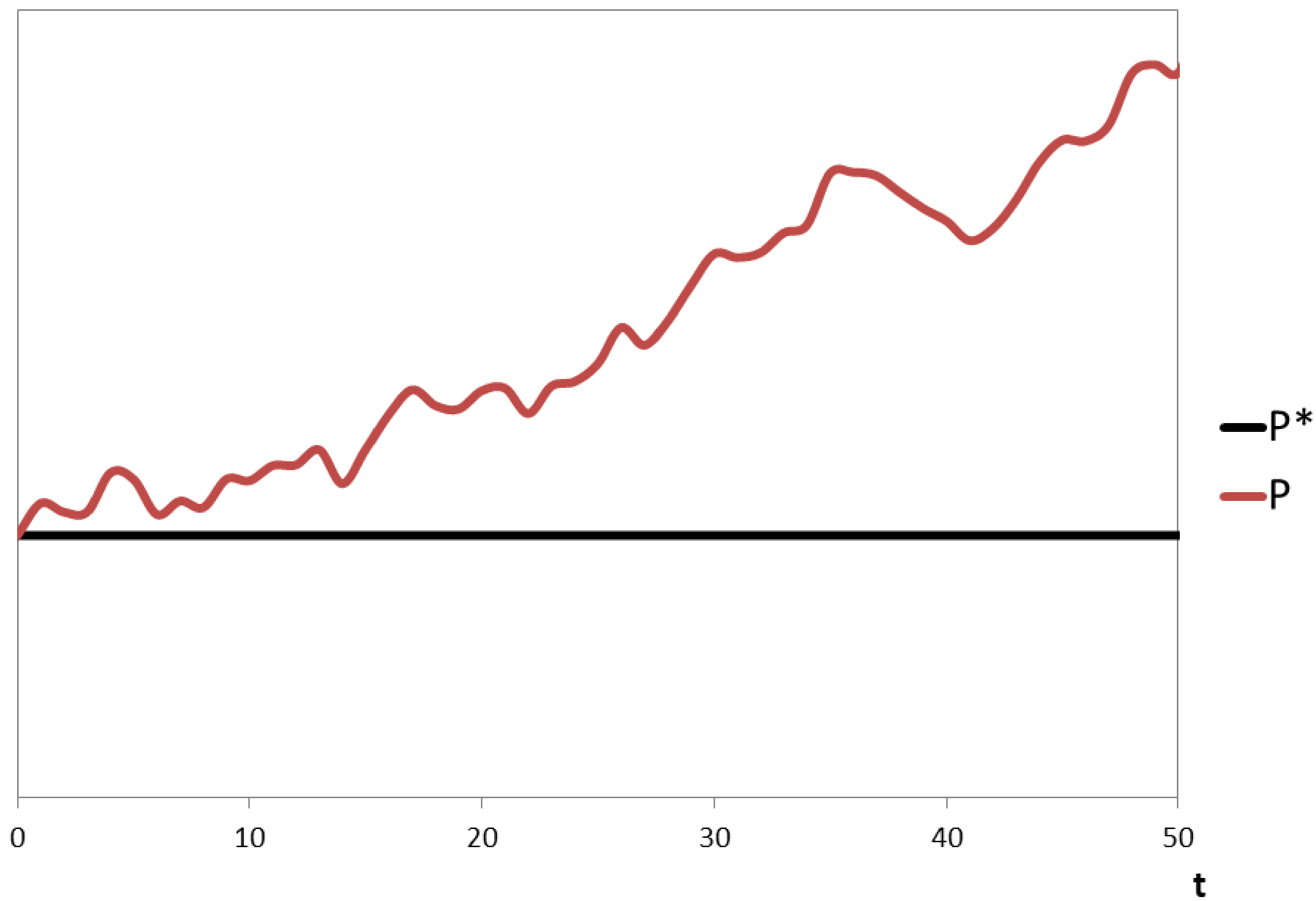
$$\begin{aligned} \mathbb{E}_t B_{t+1} &= \frac{1}{\alpha} B_t + \mathbb{E}_t \eta_{t+1} \\ &= \frac{1}{\alpha} B_t \end{aligned}$$

and then

$$P_t = P_t^* + B_t$$

is a solution with $P_t^* = \frac{1}{(1-\alpha)c} M$ and $B_t = \frac{1}{\alpha} B_{t-1} + \eta_t$.

Figure



Bursting Bubbles

Now suppose:

$$B_{t+1} = \begin{cases} \frac{\alpha}{p} B_t + \eta_{t+1} & \text{with probability } p \\ \\ 0 & \text{with probability } 1-p \end{cases}$$

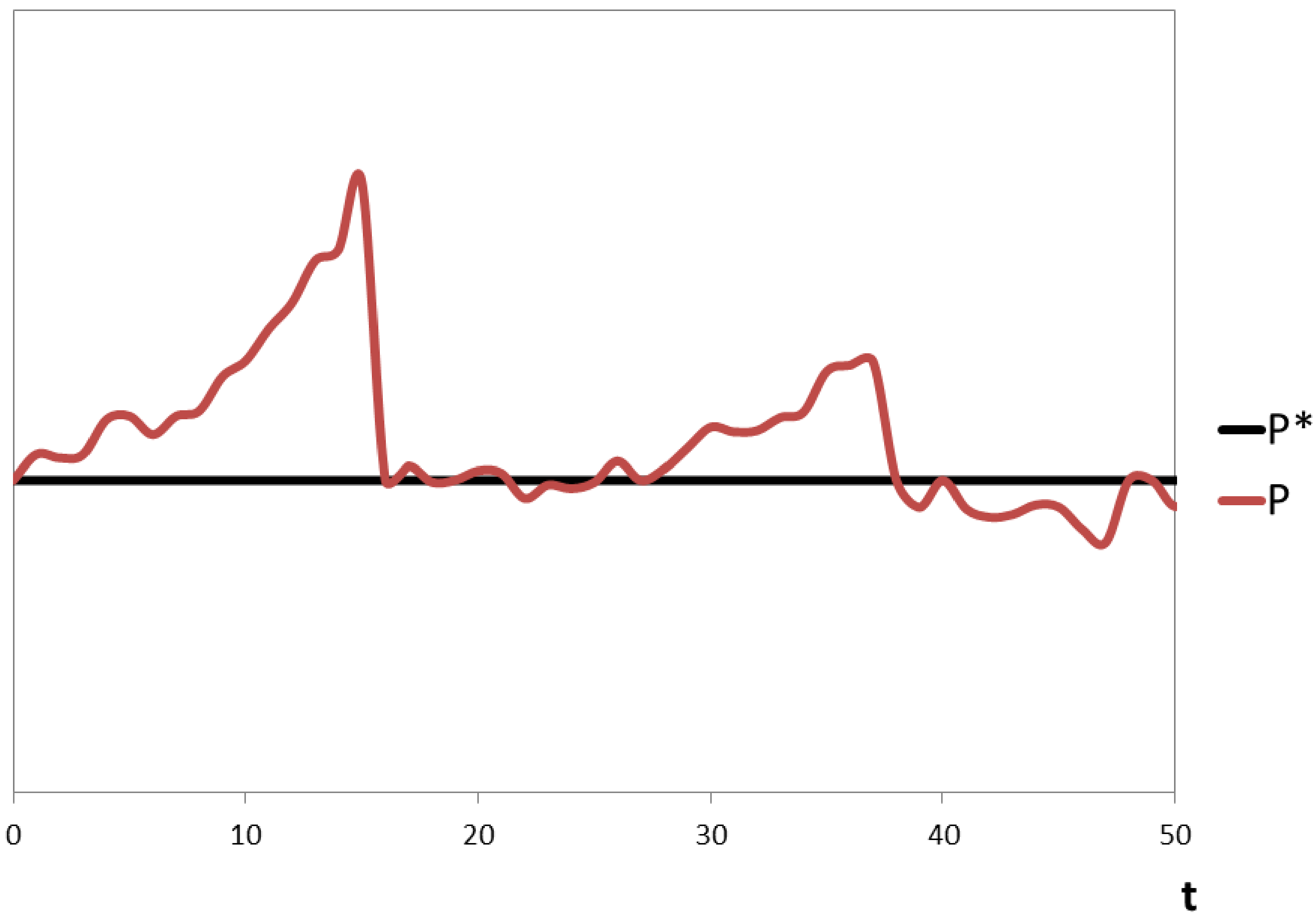
where $p \in (0,1)$ and $\{\eta_{t+1}\}$ is a zero-mean iid sequence of shocks.

From the expression above we get:

$$\begin{aligned} \mathbb{E}_t B_{t+1} &= p\mathbb{E}_t\{\frac{\alpha}{p} B_t + \eta_{t+1}\} + (1-p)\mathbb{E}_t\{0\} \\ &= p\frac{\alpha}{p} B_t + \mathbb{E}_t \eta_{t+1} \\ &= \alpha B_t \end{aligned}$$

Hence, the random variable B_t qualifies as a bubble.

Example of a bursting bubble and fundamental solution $P_t^* = \frac{1}{(1-\alpha)c} M$.



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